Bone Tissue and The Skeletal System

LEARNING OBJECTIVES

- By the end of lecture, the student should be able to:
- Define bones, and its functions.
- Classify the bones.
- Describe the endochondral and intramembranous ossification.
- Differentiate between different types of cartilage and know their locations.

THE FOUR BASIC TYPES OF TISSUE

…

- Epithelium
- Connective tissue
  - Connective tissue proper
  - Cartilage
  - Bone
  - Blood
- Muscle tissue
- Nervous tissue

- The dense, hard, and somewhat flexible connective tissue constituting the framework of the human skeleton. It is composed of compact osseous tissue surrounding spongy cancellous tissue permeated by many blood vessels and nerves and enclosed in membranous periosteum.

The Skeletal System

- Functions of the Skeletal System
  - Support against gravity
  - Leverage for muscle action - movement
  - Protection of soft internal organs
  - Blood cell production
  - Storage - calcium, phosphorous, fat

The Skeletal System

- The skeletal system includes:
  - Bones
  - Cartilages
  - Joints
  - Ligaments
  - Other connective tissues

Tissues in Bone

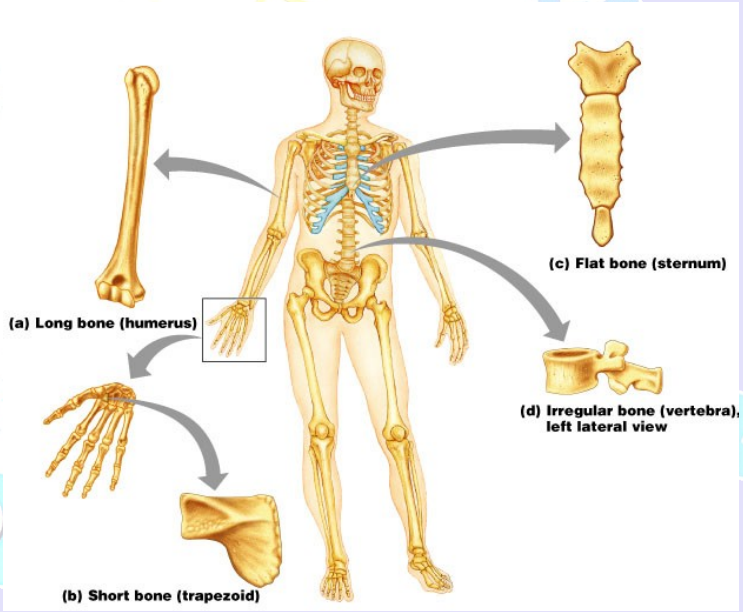
- Bones are dominated by bone tissue but also contain
  - Nervous tissue and nerves
  - Blood tissue and vessels
  - Cartilage in articular cartilages
  - Epithelial tissue lining the blood vessels

Bone (Osseous Tissue)

- Specialized cells - 2% of bone weight
- Strong flexible matrix
- Calcium phosphate crystals - two-thirds of bone weight
- Collagen fibers

Classification of bones by shape

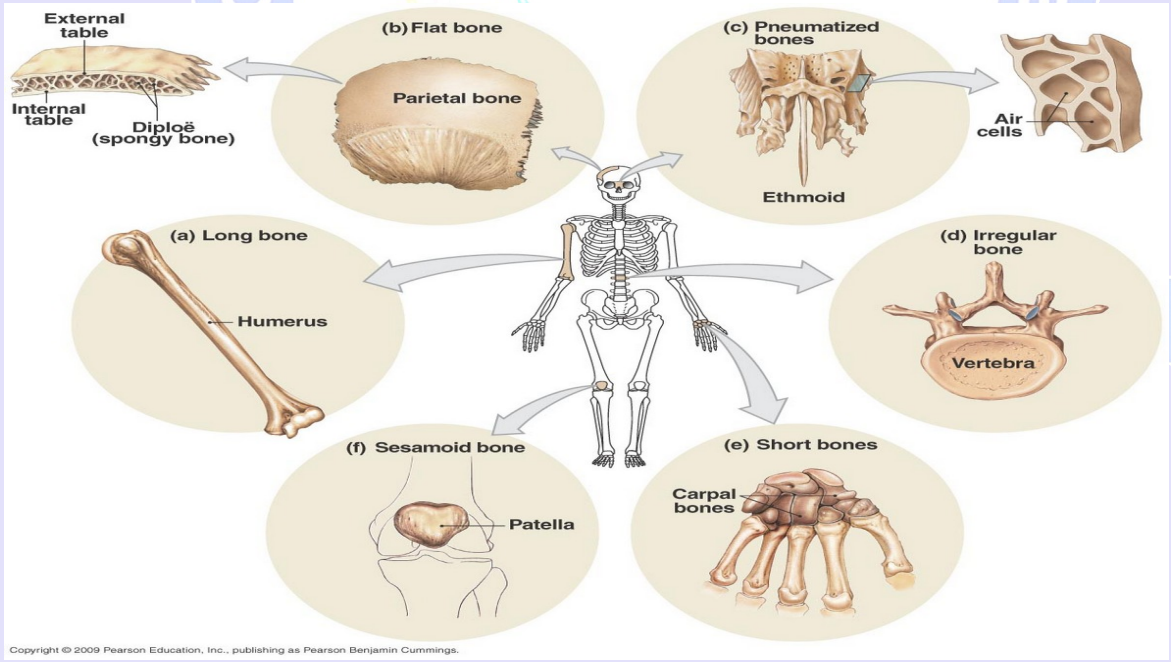
- Long bones
- Short bones
- Flat bones
- Irregular bones
- Pneumatized bones
- Sesamoid bones



(Short bones include sesmoid bones)

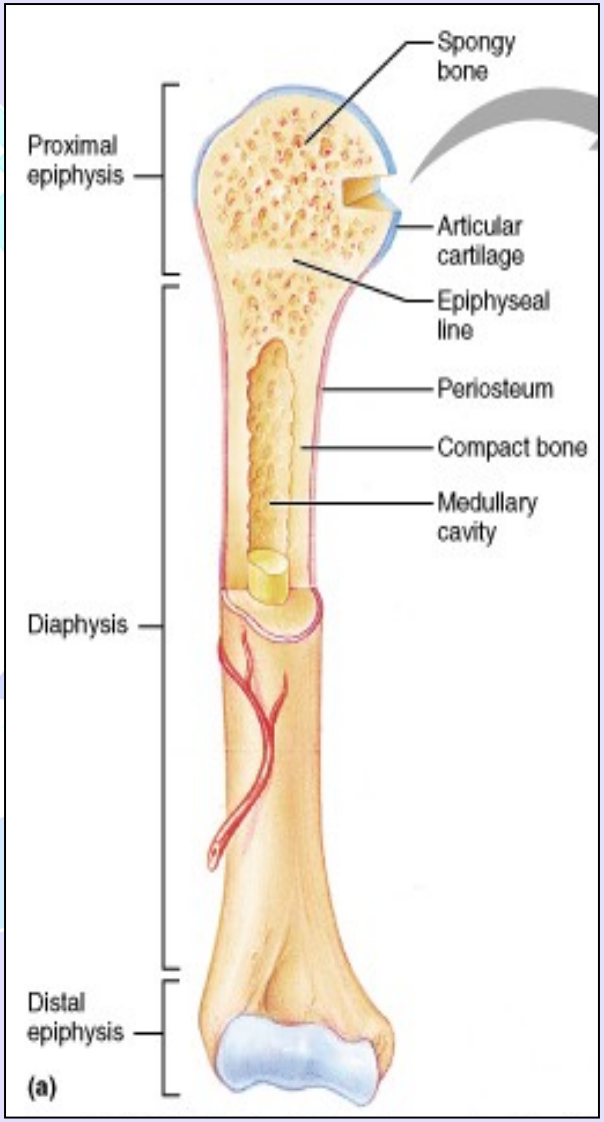
General Shapes Of Bones

- Long bones (e.g., humerus, femur)
- Short bones (e.g., carpals, tarsals, patella
- Flat bones (e.g., parietal bone, scapula, sternum)
- Irregular bones (e.g., vertebrae, hip bones)



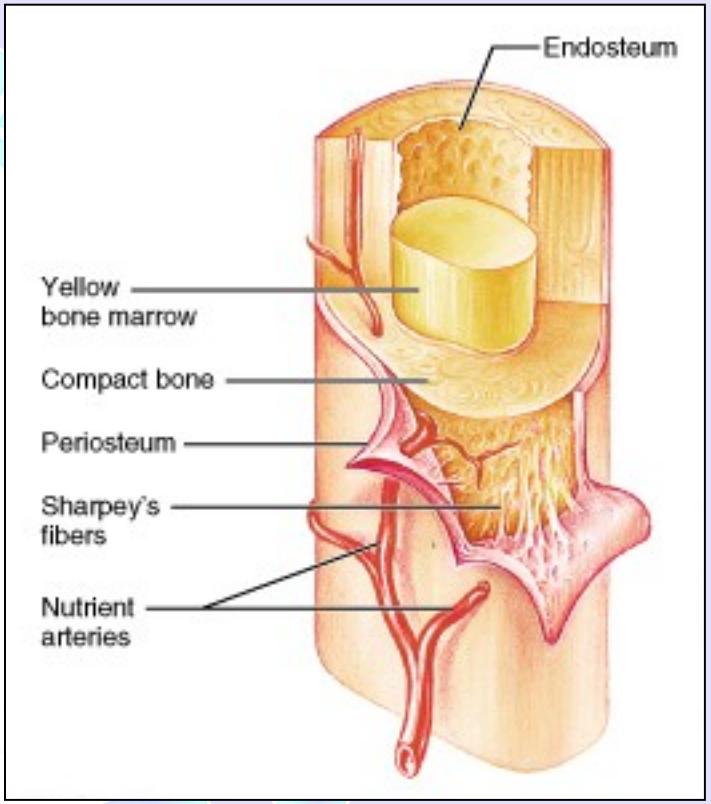
Structure of Typical Long Bone

- Diaphysis - tubular shaft forming the axis of long bones.
  - Composed of compact bone
  - Central medullary cavity
  - Contains bone marrow
- Epiphysis – expanded end of long bones.
  - Composed mostly of spongy bone
  - Joint surface is covered with *articular (hyaline) cartilage*
  - Epiphyseal lines separate the diaphysis from the epiphyses
- Metaphysis – where epiphysis and diaphysis meet



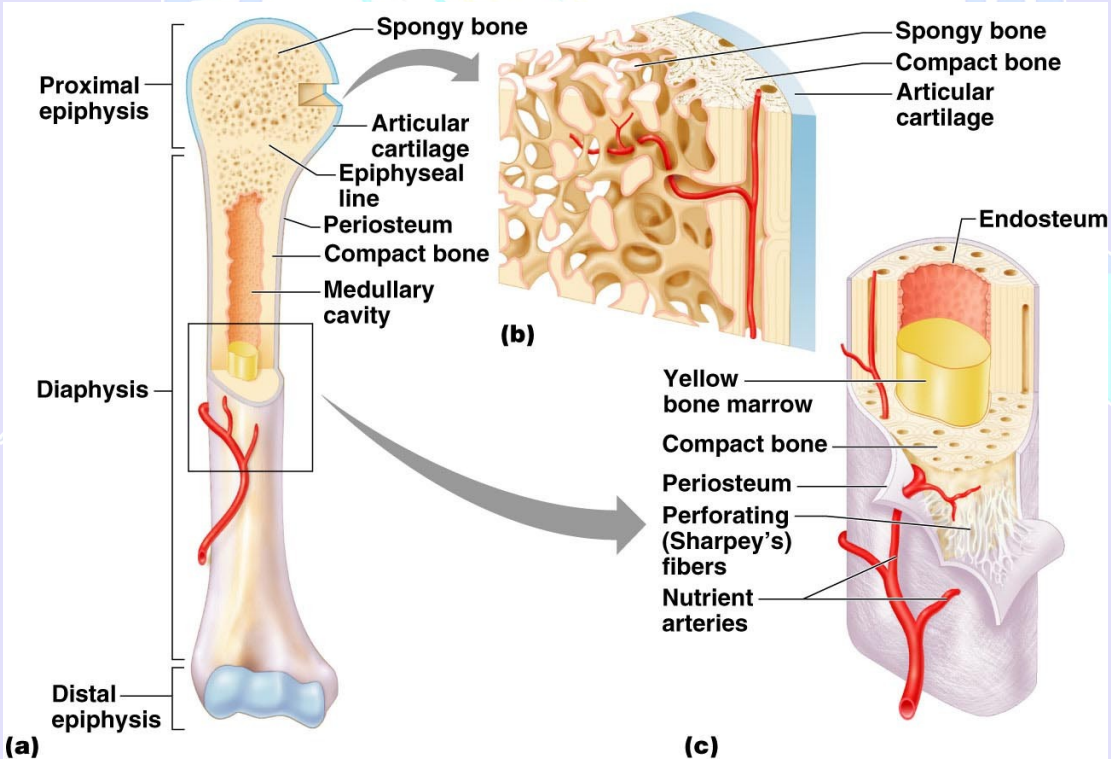
Bone Membranes

- Periosteum
  - Provides anchoring points for tendons and ligaments
  - Double-layered protective membrane, supplied with nerve fibers, blood, and lymphatic vessels entering the bone via nutrient foramina.
  - Inner osteogenic layer is composed of osteoblasts and osteoclasts
- Endosteum
  - Delicate CT membrane covering internal surfaces of bone
  - Covers trabeculae of spongy bone
  - Lines canals in compact bone
  - Also contains both osteoblasts and osteoclasts



Gross Anatomy of Bones

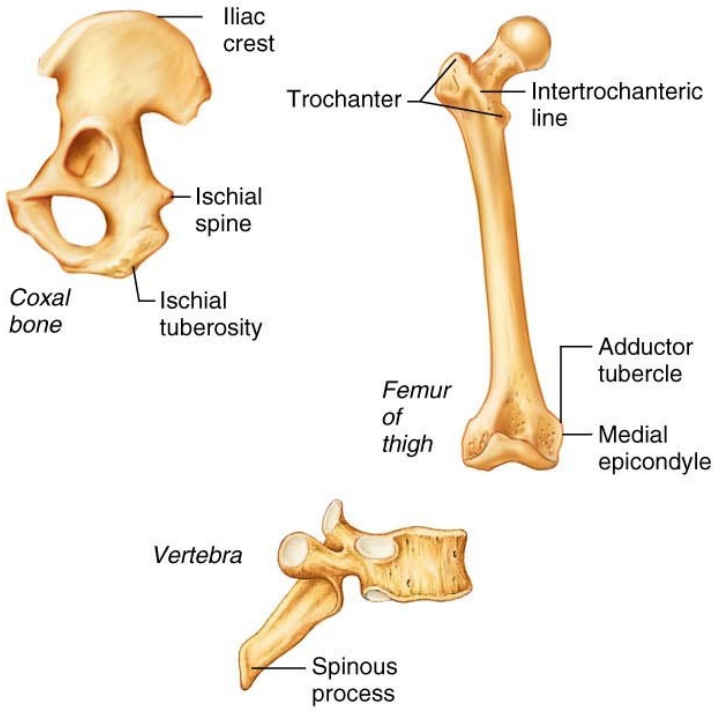
- External Features of Bones – projections, depressions, and openings that serve as sites of muscle, ligament, and tendon attachment, as joint surfaces, or conduits for blood vessels and nerves
- Compact Bone – dense outer layer
- Spongy Bone – (cancellous bone) honeycomb of trabeculae (needle-like or flat pieces) filled with bone marrow



Gross Anatomy - Bone Markings

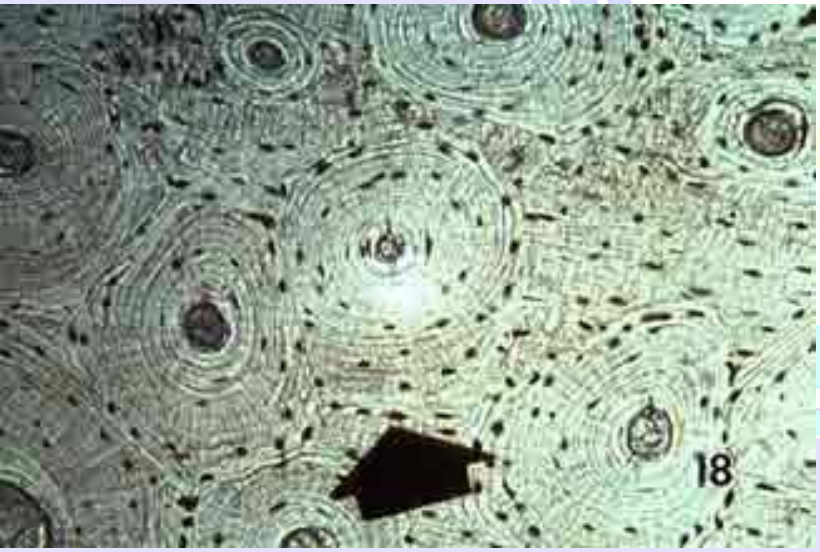
- Superficial surfaces of bones reflect stresses on them
- There are three broad categories of bone markings
  - Projections for muscle attachment
  - Surfaces that form joints
  - Depressions and openings

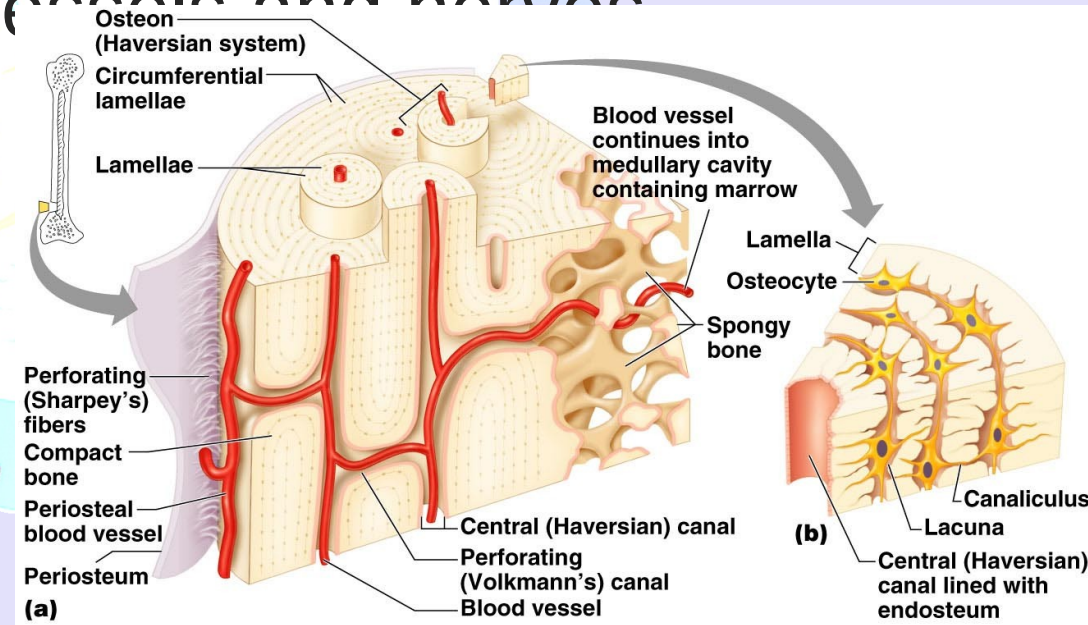
Name of Bone Marking	Description	Illustration
PROJECTIONS THAT ARE SITES OF MUSCLE AND LIGAMENT ATTACHMENT		
Tuberosity (too″be-ros′ĭ-te)	Large rounded projection; may be roughened	
Crest	Narrow ridge of bone; usually prominent	
Trochanter (tro-kan′ter)	Very large, blunt, irregularly shaped process (the only examples are on the femur)	
Line	Narrow ridge of bone; less prominent than a crest	
Tubercle (too′ber-kl)	Small rounded projection or process	
Epicondyle (ep″ĭ-kon′dīl)	Raised area on or above a condyle	
Spine	Sharp, slender, often pointed projection	
Process	Any bony prominence	



Histology of Compact Bone

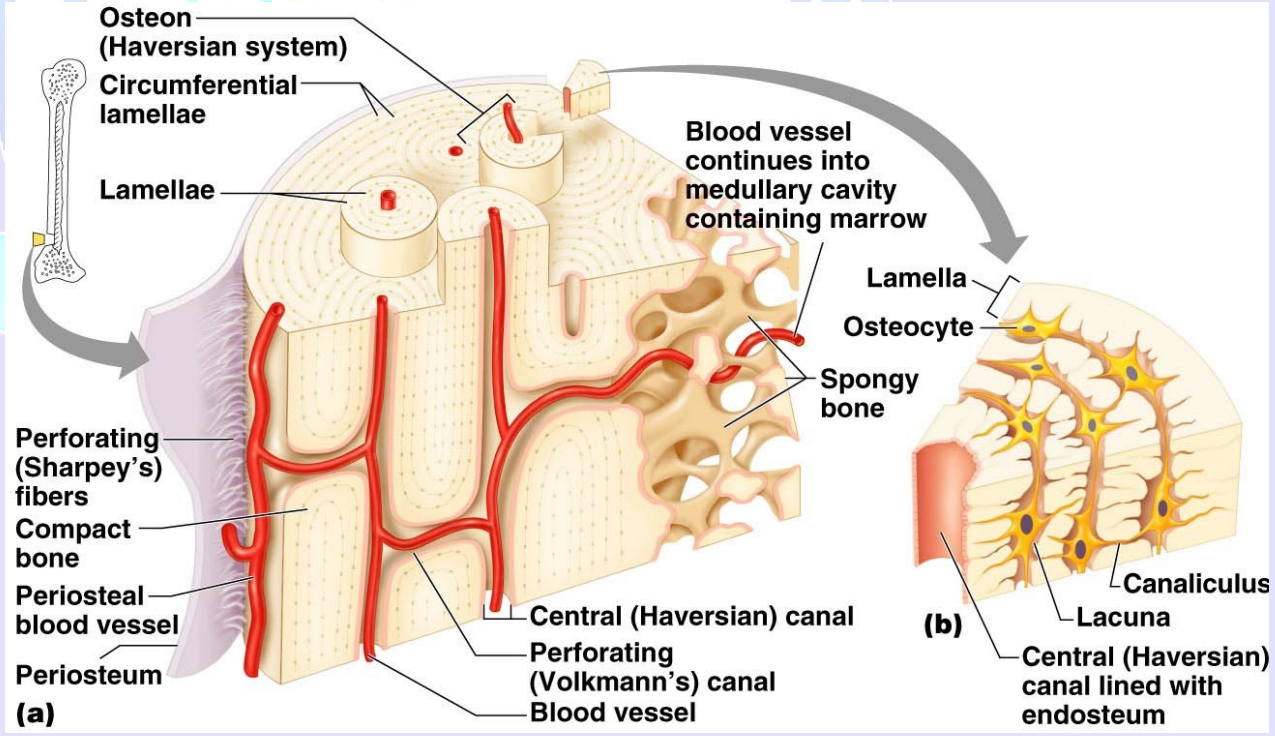
- Osteon – the structural unit of compact bone
- Lamellae – column-like matrix tubes composed of collagen and crystals of bone salts
- Central canal - (Haversian canal) canal containing blood vessels and nerves





Histology of Compact Bone

- Lacunae - cavities in bone containing osteocytes
- Canaliculi - hairlike canals that connect lacunae to each other and the central canal
- Perforating canal (Volkmann's) – channels lying at right angles to the central canal, connecting blood and nerve supply of the periosteum to the central canal

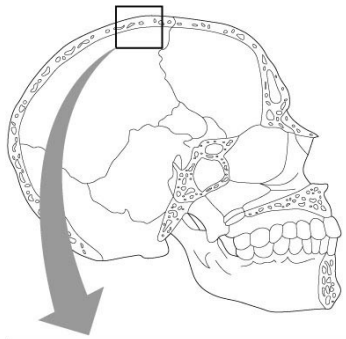


Cells in Bone

- Osteoprogenitor cells – precursors to osteoblasts
- Osteocytes - mature bone cells between lamellae
- Osteoclasts - bone-destroying cells, break down bone matrix for remodeling and release of calcium
  - Source of acid, enzymes for osteolysis
  - Calcium homeostasis
- Osteoblasts - bone-forming cells
  - Responsible for osteogenesis (new bone)
  - Source of collagen, calcium salts

The Structure of Spongy Bone

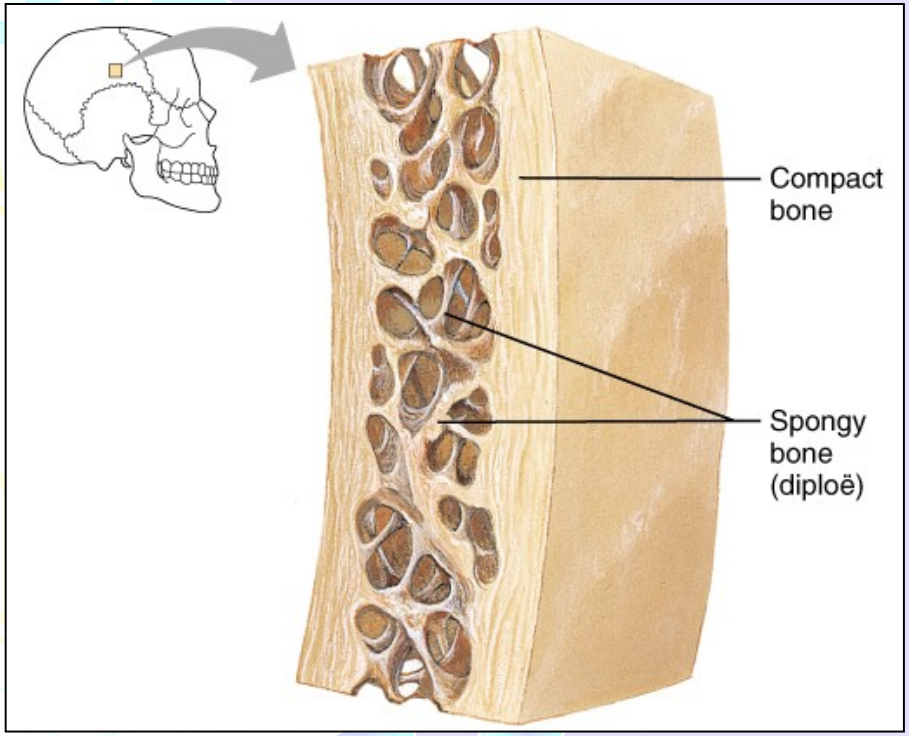
- No osteons
- Lamellae as trabeculae
  - Arches, rods, plates of bone
  - Branching network of bony tissue
  - Strong in many direction
  - Red marrow (blood forming) spaces





Short, Irregular, and Flat Bones

- Plates of periosteum-covered compact bone on the outside with endosteum-covered spongy bone, diploë, on the inside
- Have no diaphysis or epiphyses
- Contain bone marrow between the trabeculae

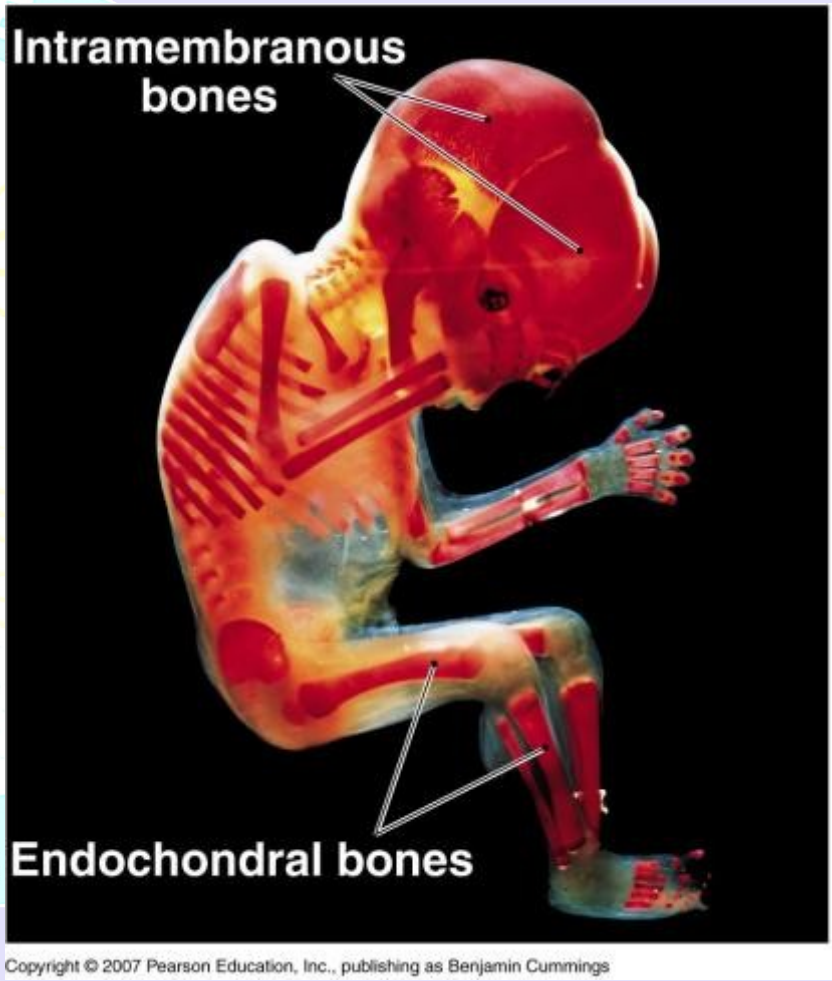


Bone Development

- Osteogenesis or Ossification – the process of bone tissue formation that leads to:
  - The formation of the skeleton in embryos
  - Bone growth until early adulthood
  - Bone thickness, remodeling, and repair

Formation of the Skeleton

- Before week 8, the skeleton of a human embryo consists of fibrous membanes and hyaline cartilage
- Intramembranous ossification – bone develops from a fibrous connective tissue membrane. The flat bones of the skull (frontal, parietal, temporal, occipital) and the clavicles are formed this way.
- Endochondral ossification – bone forms by replacing hyaline cartilage, uses hyaline cartilage “bones” as patterns



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Bone Formation in 16-Week-Old Fetus

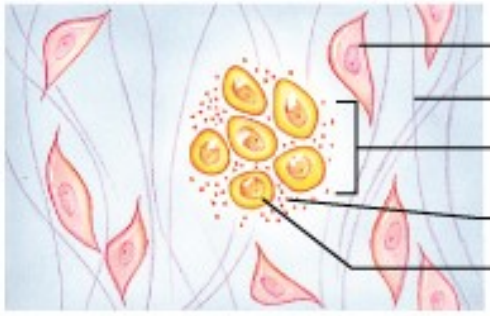
Bone Formation and Growth

- Intramembranous Ossification
  - Ossification—Process of converting other tissues to bone
  - Forms flat bones of skull, mandible, clavicle
  - Stem cells differentiate to osteoblasts
  - Produces spongy bone, then compact bone

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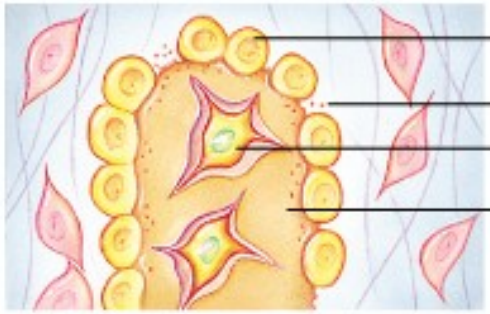
Intramembranous Ossification

- An ossification center appears in the fibrous connective tissue membrane
- Osteoblasts secrete bone matrix within the fibrous membrane
- Osteoblasts mature into osteocytes



① **An ossification center appears in the fibrous connective tissue membrane.**

- Selected centrally located mesenchymal cells cluster and differentiate into osteoblasts, forming an ossification center.



② **Bone matrix (osteoid) is secreted within the fibrous membrane.**

- Osteoblasts begin to secrete osteoid, which is mineralized within a few days.
- Trapped osteoblasts become osteocytes.

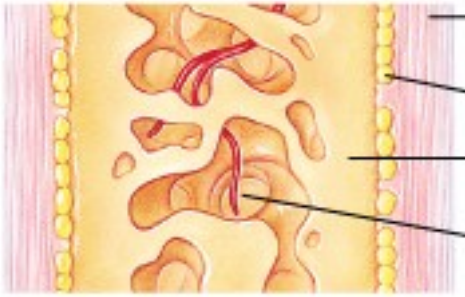
Intramembranous Ossification

- The bone matrix develops into trabeculae.
- The trabeculae formed from various ossification centers fuse with one another to create spongy bone.
- Eventually the spaces between trabeculae fill with red bone marrow.



③ **Woven bone and periosteum form.**

- Accumulating osteoid is laid down between embryonic blood vessels, which form a random network. The result is a network (instead of lamellae) of trabeculae (woven bone).
- Vascularized mesenchyme condenses on the external face of the woven bone and becomes the periosteum.



④ **Bone collar of compact bone forms and red marrow appears.**

- Trabeculae just deep to the periosteum thicken, forming a woven bone collar that is later replaced with mature lamellar bone.
- Spongy bone (diploë), consisting of distinct trabeculae, persists internally and its vascular tissue becomes red marrow.

Bone Formation and Growth

- Endochondral Ossification
  - Most bones formed this way
  - Cartilage model replaced by bone
  - Replacement begins in middle (diaphysis)
  - Replacement follows in ends (epiphyses)

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Endochondral Ossification

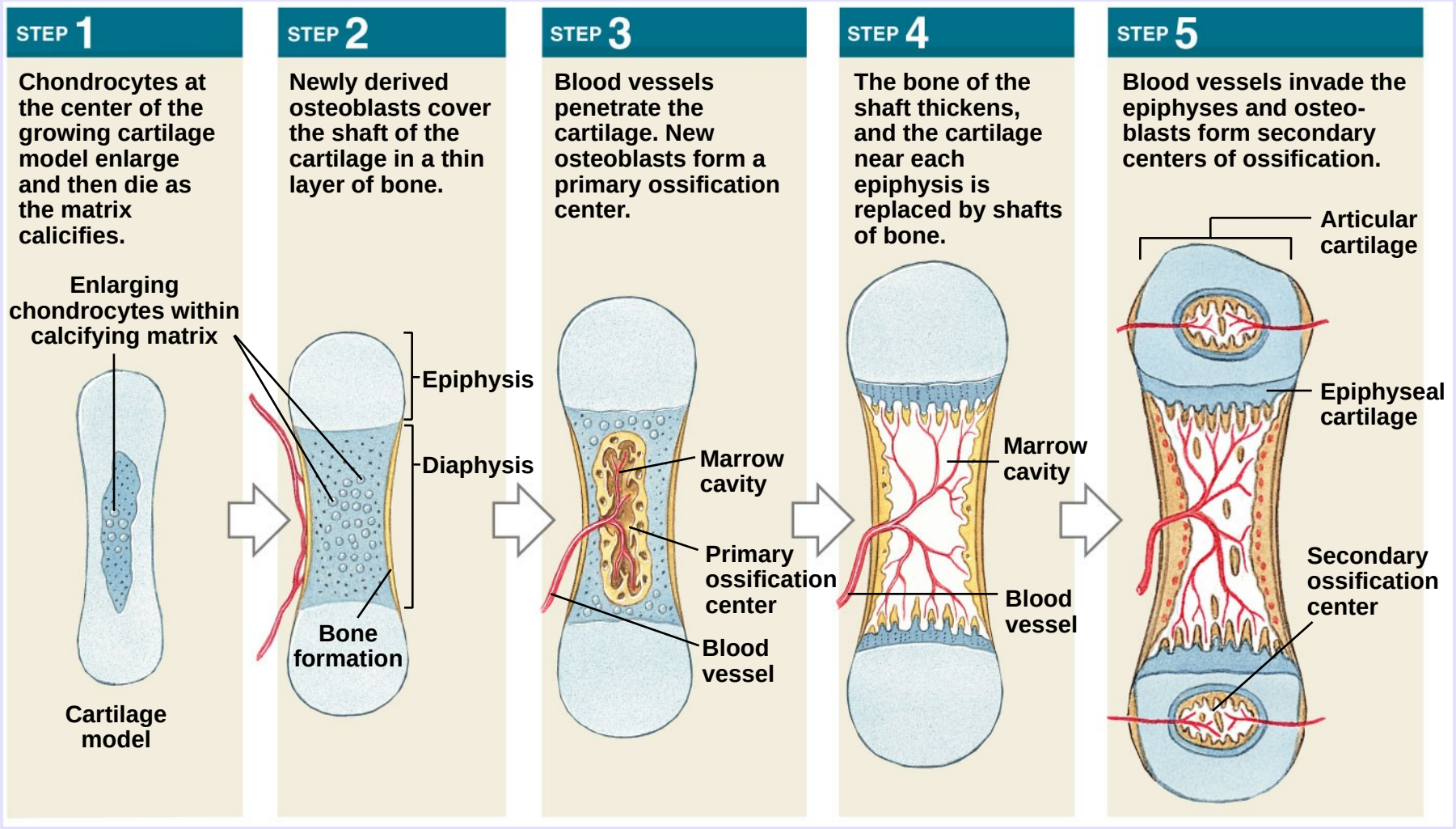
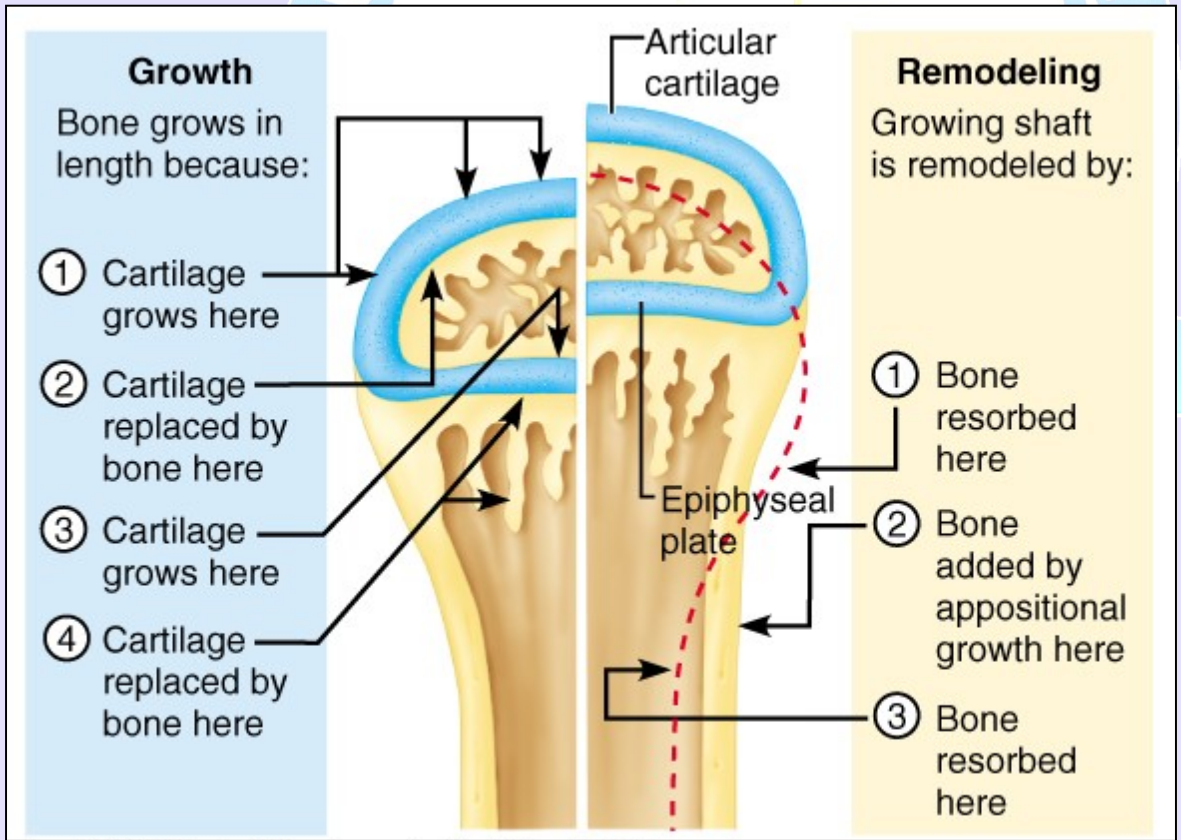


Figure 6-5
1 of 6

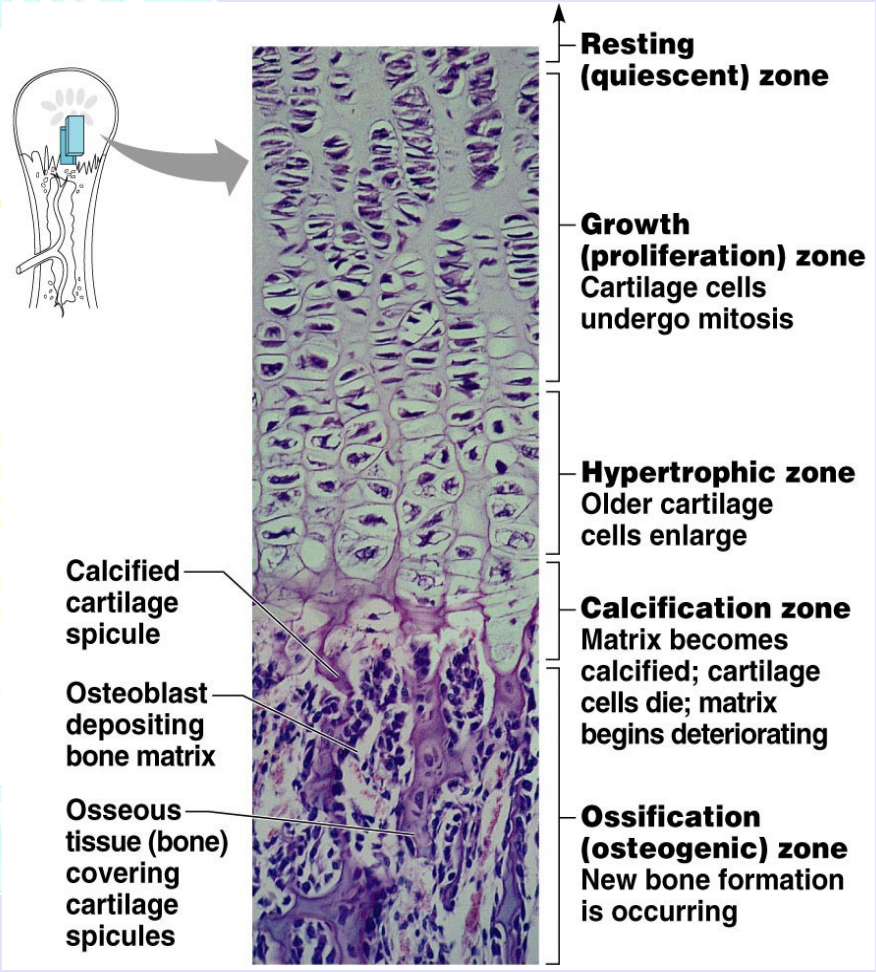
Longitudinal Bone Growth

- Longitudinal Growth (interstitial) – cartilage continually grows and is replaced by bone
  - Bones lengthen entirely by growth of the **epiphyseal plates**
  - Cartilage is replaced with bone CT as quickly as it grows
  - *Epiphyseal plate maintains constant thickness*



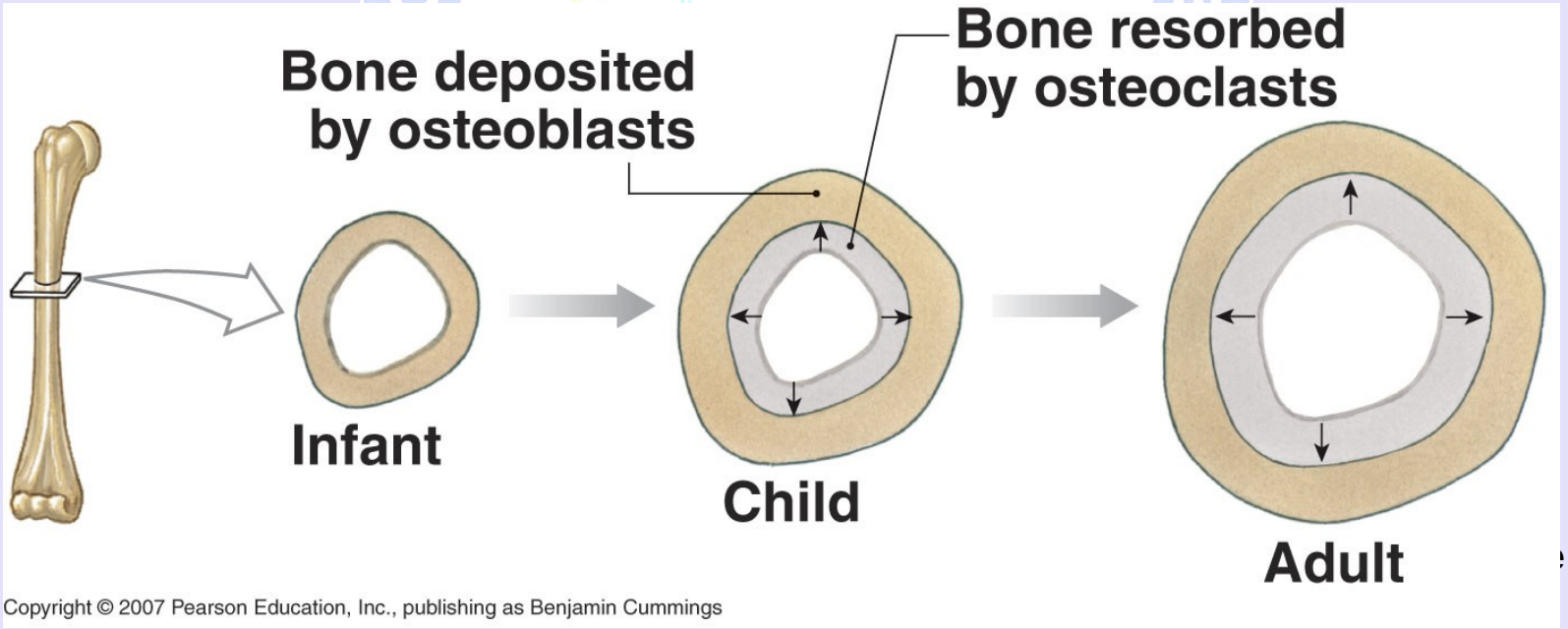
Epiphyseal Plate

- Cartilage is organized for quick, efficient growth
- Cartilage cells form tall stacks
  - **Chondroblasts** at the top of stacks divide quickly
  - Pushes the epiphysis away from the diaphysis
  - Lengthens entire long bone
- Older chondrocytes signal surrounding matrix to calcify, then die and disintegrate
  - Leaves long **trabeculae** (spicules) of calcified cartilage on diaphysis side
  - **Trabeculae** are partly eroded by osteoclasts
  - **Osteoblasts** then cover trabeculae with bone tissue
  - Trabeculae finally eaten away from their tips by **osteoclasts**



Appositional Bone Growth

- Growing bones widen as they lengthen
- Appositional growth – growth of a bone by addition of bone tissue to its surface
- Bone is resorbed at endosteal surface and added at periosteal surface
  - Osteoblasts – add bone tissue to the external surface of the diaphysis
  - Osteoclasts – remove bone from the internal surface of the diaphysis



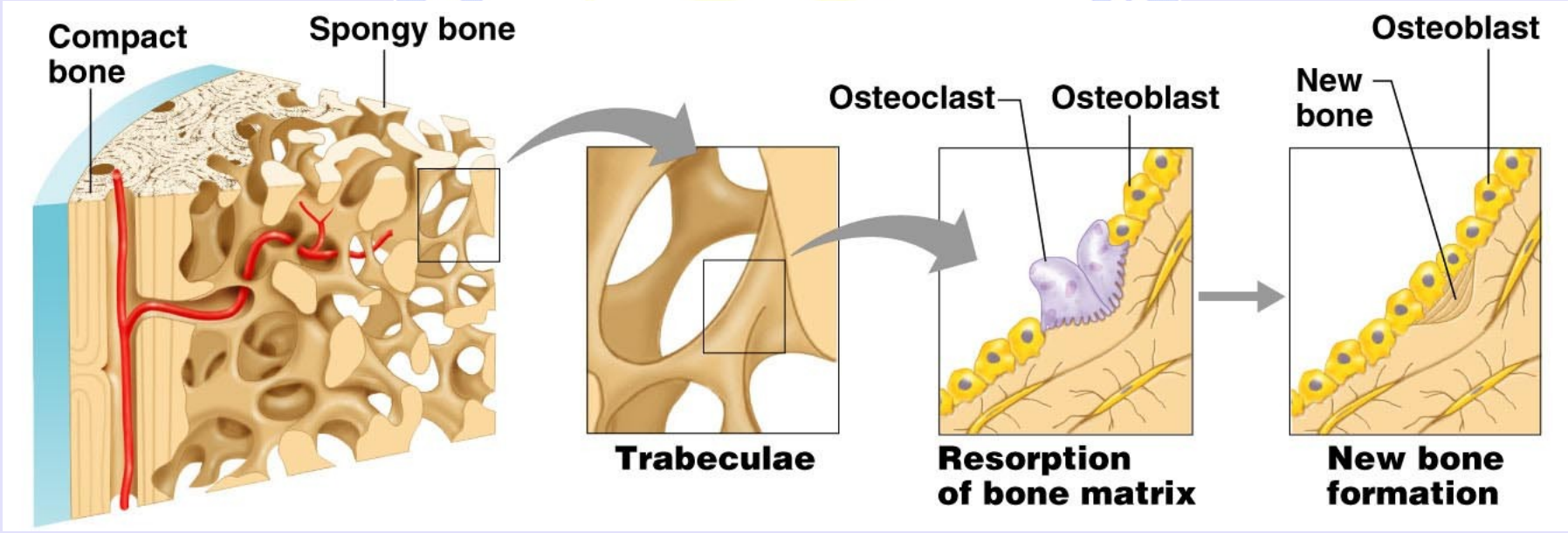
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Bone - Remodeling/Homeostasis

- Role of Remodeling in Support
  - Remodeling—Continuous breakdown and reforming of bone tissue
  - Shapes reflect applied loads
  - Mineral turnover enables adapting to new stresses
- What you don’t use, you lose. The stresses applied to bones during exercise are essential to maintaining bone strength and bone mass

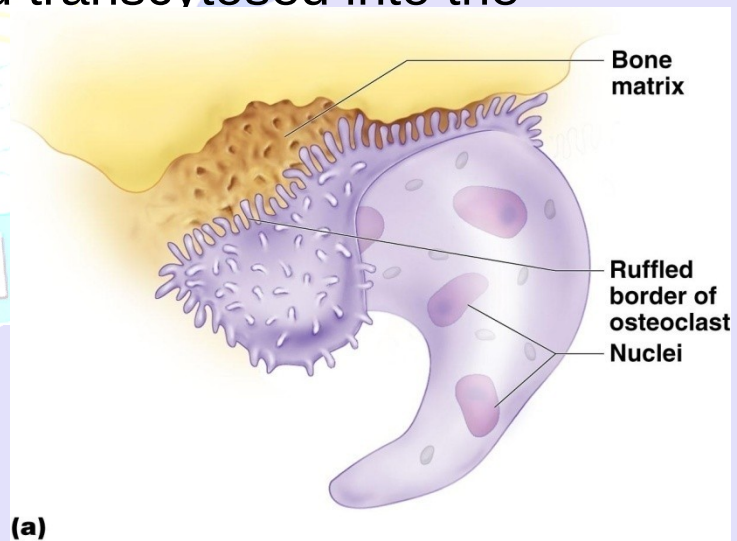
Bone Remodeling

- Bone is active tissue – small changes in bone architecture occur continuously – 5 to 7% of bone mass is recycled weekly – spongy bone is replaced every 3-4 years and compact bone approximately every 10 years
- Remodeling Units – adjacent osteoblasts and osteoclasts deposit and reabsorb bone at periosteal and endosteal surfaces

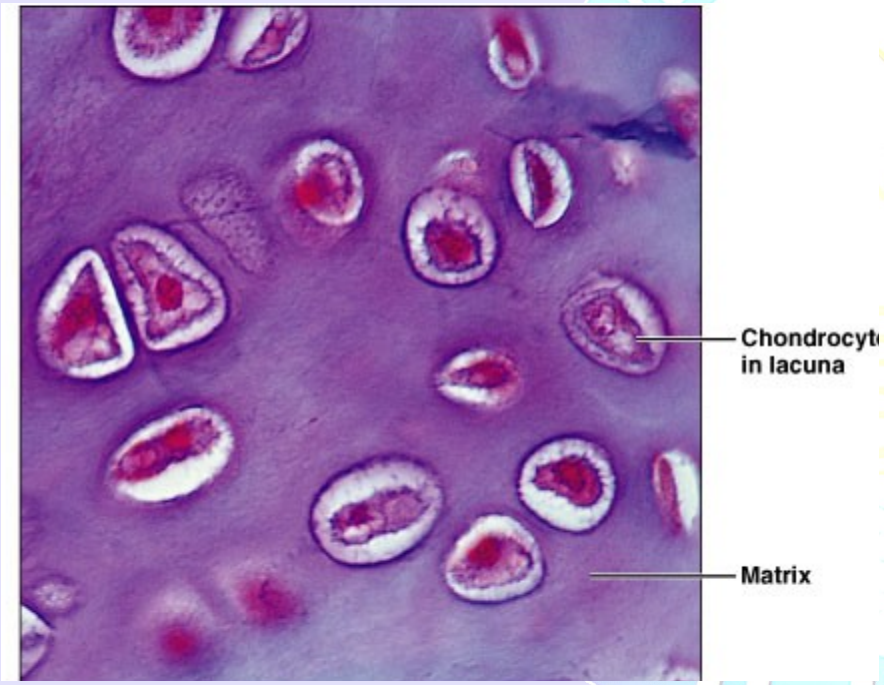


Bone Remodeling

- Bone Depostition
  - Occurs when bone is injured or extra strength is needed
  - Requires a healthy diet - protein, vitamins C, D, and A, and minerals (calcium, phosphorus, magnesium, manganese, etc.)
- Bone Resorption
  - Accomplished by Osteoclasts (multinucleate phagocytic cells)
  - Resorption involves osteoclast secretion of:
    - Lysosomal enzymes that digest organic matrix
    - HCl that converts calcium salts into soluble forms
  - Dissolved matrix is endocytosed and transcytosed into the interstitial fluid → the blood



Cartilage is connective tissue



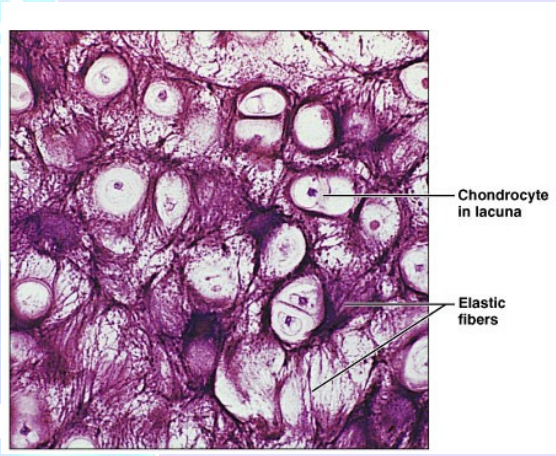
(hyaline cartilage)

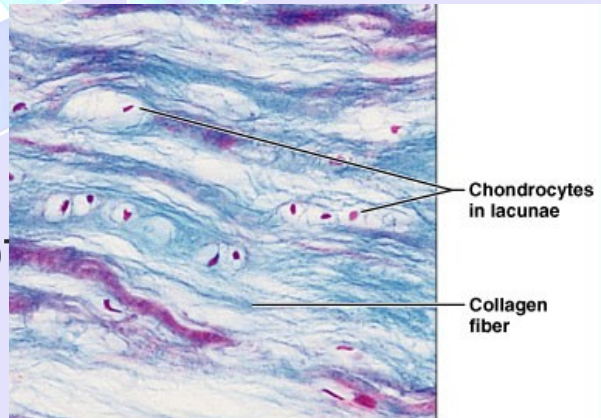
- Cells called **chondrocytes**
- Abundant **extracellular matrix**
  - Fibers: collagen & elastin
  - Jellylike ground substance of complex sugar molecules
  - 60-80% water (responsible for the resilience)
  - No nerves or vessel

Types of cartilage

1. Hyaline cartilage: flexible and resilient
   - Chondrocytes appear spherical
   - *Lacuna* – cavity in matrix holding chondrocyte
   - Collagen the only fiber
2. Elastic cartilage: highly bendable
   - Matrix with elastic as well as collagen fibers
   - Epiglottis, larynx and outer ear
3. Fibrocartilage: resists compression and tension
   - Rows of thick collagen fibers alternating with rows of chondrocytes (in matrix)
   - Knee menisci and annunulus fibrosis o intervertebral discs







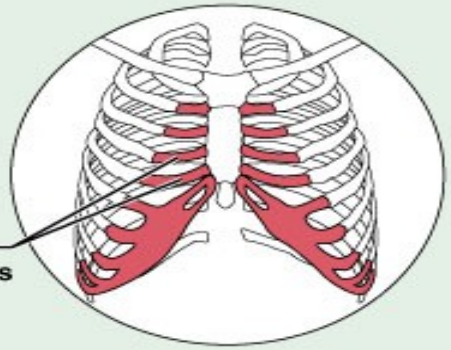
Hyaline Cartilage

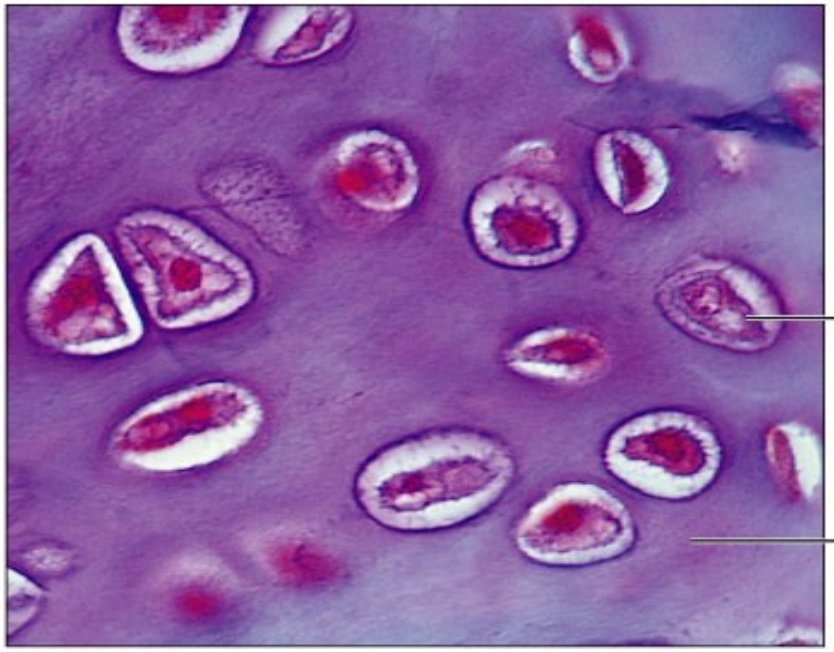
(g) Cartilage: hyaline

Description: Amorphous but firm matrix; collagen fibers form an imperceptible network; chondroblasts produce the matrix and when mature (chondrocytes) lie in lacunae.

Function: Supports and reinforces; has resilient cushioning properties; resists compressive stress.

Location: Forms most of the embryonic skeleton; covers the ends of long bones in joint cavities; forms costal cartilages of the ribs; cartilages of the nose, trachea, and larynx.





Photomicrograph: Hyaline cartilage from the trachea (300×).

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Elastic Cartilage

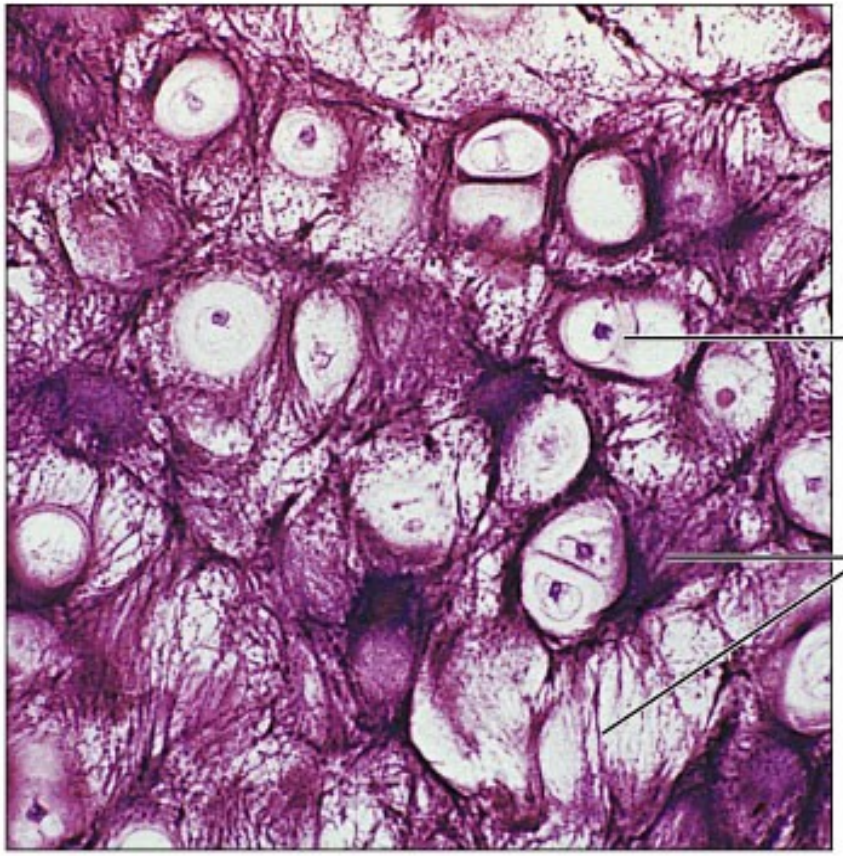
(h) Cartilage: elastic

Description: Similar to hyaline cartilage, but more elastic fibers in matrix.

Function: Maintains the shape of a structure while allowing great flexibility.

Location: Supports the external ear (pinna); epiglottis.





Photomicrograph: Elastic cartilage from the human ear pinna; forms the flexible skeleton of the ear (400×).

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Fibrocartilage

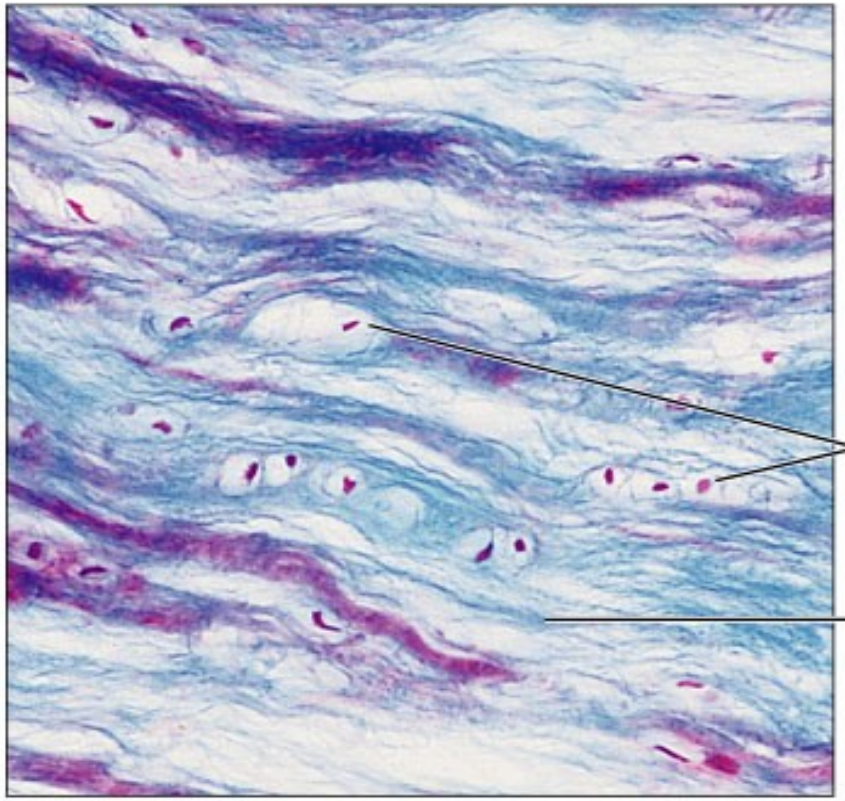
(i) Cartilage: fibrocartilage

Description: Matrix similar to but less firm than that in hyaline cartilage; thick collagen fibers predominate.

Function: Tensile strength with the ability to absorb compressive shock.

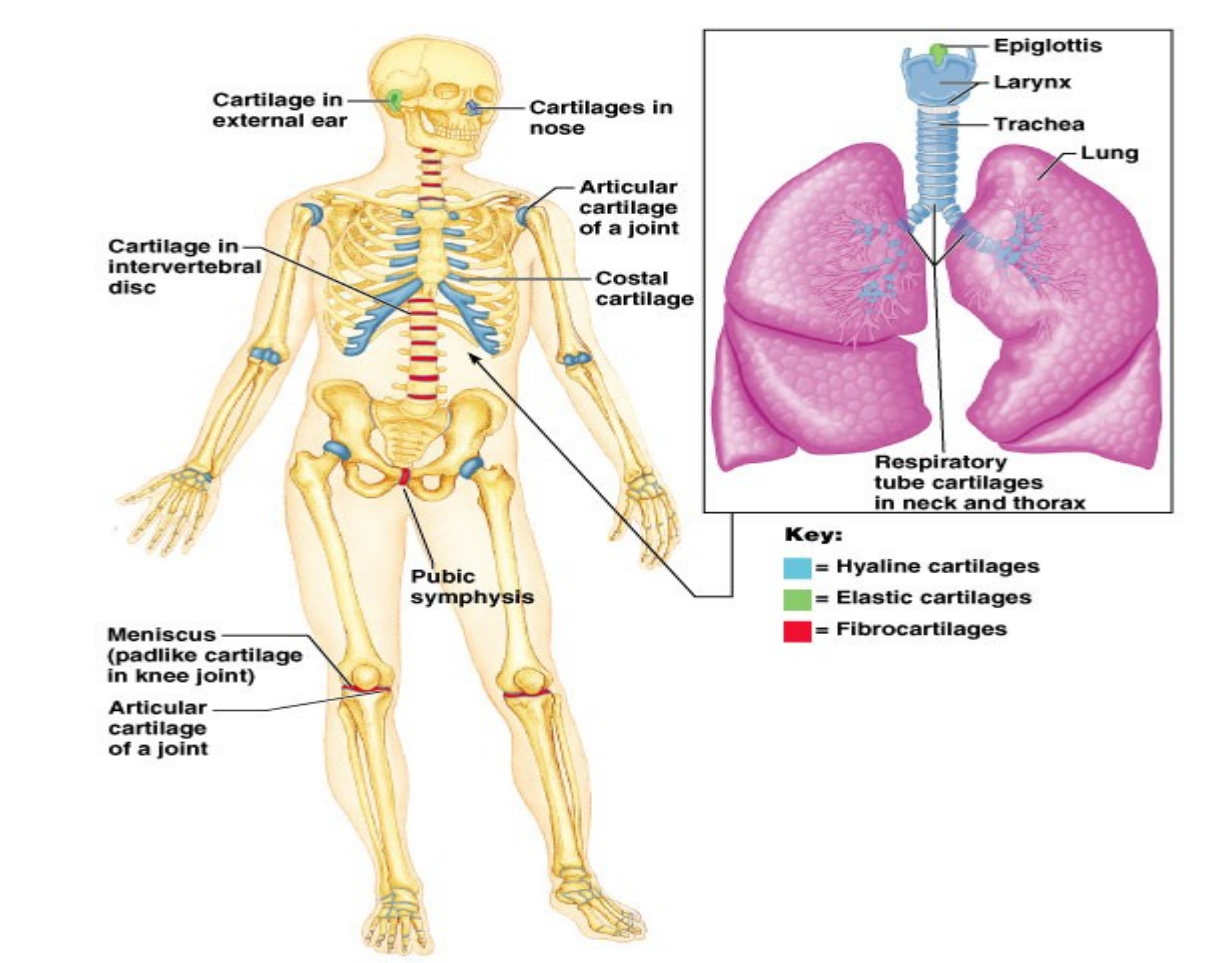
Location: Intervertebral discs; pubic symphysis; discs of knee joint.





Photomicrograph: Fibrocartilage of an intervertebral disc (200×).

Locations of the different kinds of cartilage



Growth of cartilage

- Appositional
  - “Growth from outside”
  - Chrondroblasts in perichondrium (external covering of cartilage) secrete matrix
- Interstitial
  - “Growth from within”
  - Chondrocytes within divide and secrete new matrix
- Cartilage stops growing in late teens (chrondrocytes stop dividing)
- Regenerates poorly in adults

Disorders of cartilage and bone

- Defective collagen
  - Numerous genetic disorders
  - eg. Osteogenesis imperfecta (brittle bones) – AD (autosomal dominant)
  - eg. Ehlers-Danlos (rubber man)
- Defective endochondral ossification
  - eg. Achondroplasia (short –limb dwarfism) - AD
- Inadequate calcification (requires calcium and vitamin D)
  - Osteomalacia (soft bones) in adults
  - Rickets in children

Note: "AD" here means autosomal dominant

- Pagets disease – excessive turnover, abnormal bone
- Osteosarcoma – bone cancer, affecting children primarily
- Osteoporosis – usually age related, esp. females
  - Low bone mass and increased fractures
  - Resorption outpaces bone deposition

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THANKS
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